

CSC495: Automated Detection of Volcanic Unrest Using Machine Learning on RCM InSAR Data

Final Report

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Abstract

This report presents the development and evaluation of a machine learning-based system for detecting volcanic unrest using RADARSAT Constellation Mission (RCM) InSAR data. The project applies pre-trained convolutional neural networks (CNNs) to wrapped interferograms and assesses performance across different deformation intensities and volcanic sites. Sensitivity analysis was conducted to estimate detection thresholds, and a scalable inference workflow was built using AWS Batch and Docker. Results show that the models can identify deformation events, particularly at higher deformation levels, and that spatial coherence improves when using Universal Transverse Mercator (UTM) coordinates. While limitations remain—particularly in detecting subtle deformation—the system demonstrates potential for large-scale, near real-time monitoring of volcanic unrest. The study demonstrates the potential for automating volcanic unrest detection on a large scale.

Introduction

Background

Volcanoes are often associated with hotspots like Iceland, Indonesia, or Hawaii, but they're actually distributed across the globe — including in Canada. In fact, Canada is home to 28 volcanic fields, primarily located in British Columbia and the Yukon. While these regions may seem remote, the impacts of a volcanic eruption can extend far beyond the immediate area. Volcanic ash, for instance, can travel thousands of kilometers through the atmosphere, disrupting air traffic, damaging infrastructure, and posing serious health risks to communities far from the eruption source.

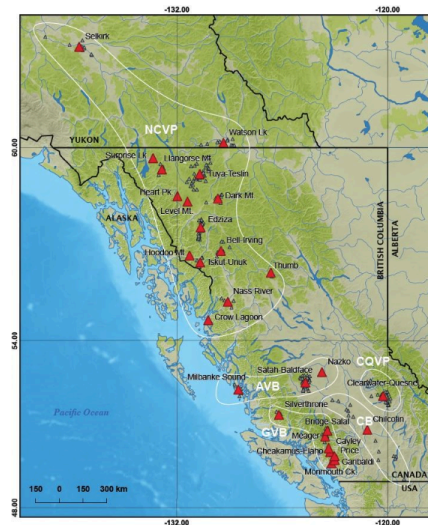


Figure 1: Volcanoes in Canada that have erupted within the past 1.8 million years. Red triangles represent volcanic complexes or fields with assessed hazard levels, while smaller grey triangles indicate individual vents within those complexes.

Earth observation missions like the [RADARSAT Constellation Mission \(RCM\)](#) help to better monitor such hazards. Launched on June 12, 2019, RCM consists of three identical satellites working in tandem to deliver high-resolution Synthetic Aperture Radar (SAR) data. This data is particularly valuable for observing ground changes in remote or cloud-covered regions. One of the most powerful applications of SAR data is Interferometric Synthetic Aperture Radar

(InSAR) — a technique that enables precise monitoring of ground deformation, a key indicator of volcanic unrest.

Volcanic deformation, which results from magma movement beneath the surface, has been statistically linked to eruptions. Unlike many other satellite-based methods, InSAR can often detect these deformations *before* an eruption occurs, offering critical lead time for early warning systems. InSAR's ability to identify subtle ground movements from space makes it especially valuable for monitoring large, remote, or under-instrumented volcanic regions — including those in developing countries where on-the-ground equipment and expertise may be limited.

InSAR works by combining two or more SAR images of the same area taken at different times. By analyzing the phase difference between these images, scientists can detect tiny ground movements, such as subsidence, uplift, or horizontal motion. The result is an interferogram — a color-coded map that visualizes ground displacement over time. These insights are critical for detecting the early signs of volcanic unrest.

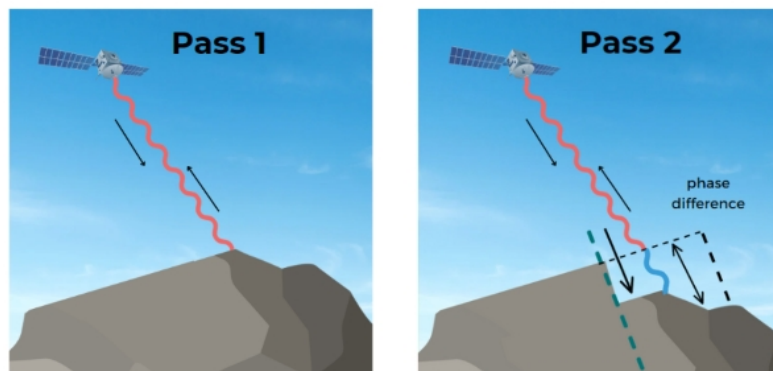


Figure 2: InSAR detects subtle ground movements—such as uplift, subsidence, and horizontal shifts—by comparing radar images of the same area captured at different times.

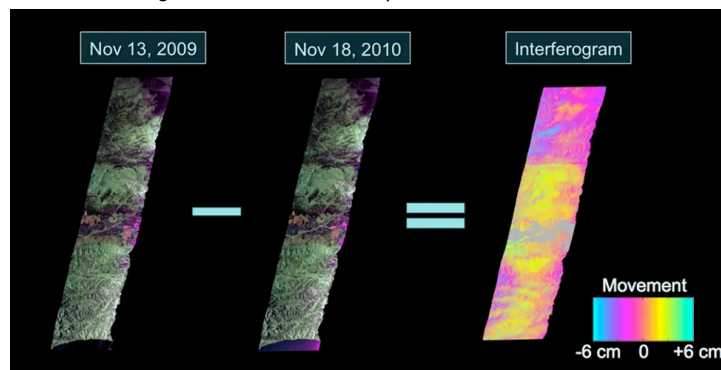


Figure 3: Example interferogram from NASA's UAVSAR over the San Andreas Fault in California. The fault is visible in the upper half where pink and yellow fringes meet, indicating surface movement between the two radar acquisitions.

Traditionally, volcanic deformation was monitored using ground-based instruments like GPS stations and tiltmeters. While still essential, these point-based tools are now enhanced by remote sensing methods like InSAR, which provide broader spatial coverage and the ability to

monitor otherwise inaccessible areas. This integration enables researchers to capture early, subtle signs of volcanic activity that might otherwise go undetected.

However, modern satellites such as Sentinel-1 and RCM generate enormous volumes of SAR imagery with high temporal and spatial resolution. While this abundance of data is a scientific goldmine, it poses a significant challenge: manual inspection and interpretation are no longer feasible at scale.

This is where Machine Learning (ML) comes into play. ML algorithms can be trained to automatically identify patterns of ground deformation in vast SAR datasets, enabling rapid and scalable analysis. This facilitates near real-time monitoring — the ability to detect and respond to surface changes within hours or days of satellite data acquisition, rather than weeks or months. Such timely insights are crucial for informing decision-makers, activating early warning systems, and mitigating the risks posed by volcanic hazards — especially in data-scarce environments.

This project aims to build upon previous work in ML-based volcanic unrest detection, particularly focusing on InSAR data. We broke this project into three major milestones to structure our progress and ensure a clear development path:

- **Milestone 1: Literature Review, Data Acquisition & Baseline Model Implementation**

We began by reviewing existing literature on machine learning approaches to volcanic unrest detection and the characteristics of InSAR data. We then acquired and preprocessed relevant InSAR datasets from the Global Volcano Watch archive. A pre-trained convolutional neural network (CNN) was adapted and fine-tuned for our specific use case, with preliminary testing performed on a subset of volcanoes known to exhibit deformation events.

- **Milestone 2: Sensitivity Analysis & Model Optimization**

In this phase, we conducted a quantitative sensitivity analysis to assess how detectable various levels of ground deformation are across different volcanoes. We focused on understanding the model's detection threshold and improving its accuracy by refining parameters and evaluating performance under different conditions of volcanic activity.

- **Milestone 3: Workflow Development**

The final milestone involves building a robust and scalable pipeline capable of performing real-time inference on incoming RCM SAR data. This includes implementing real-time detection capabilities, exploring alternative machine learning frameworks, and using synthetic data to support future model retraining. The project concludes with a comprehensive evaluation of the system's performance and recommendations for future improvements.

Problem Statement

Given the growing availability of satellite data, there is an urgent need to automate the detection of volcanic unrest signals. This project focuses on utilizing pre-trained CNN models on RCM InSAR data to detect ground deformation related to volcanic activity.

Objectives

- Evaluate the performance of pre-trained CNNs for detecting volcanic unrest signals.
- Conduct a sensitivity analysis across a range of volcanoes to quantify how much vertical ground deformation (uplift or subsidence) is typically associated with magmatic unrest, helping to define deformation thresholds which can improve detection and monitoring.
- Develop a scalable workflow for near real-time inference and detection as new data becomes available.

Scope

The project focuses on detecting volcanic unrest in InSAR data acquired by the [RADARSAT Constellation Mission](#). It evaluates the generalization of pre-trained CNN models to real-world data and addresses challenges such as data preprocessing and model adaptation.

Literature Review

Previous Work

The literature on machine learning approaches for volcanic deformation detection using InSAR data is extensive. Key papers reviewed include:

1. Anantrasirichai, N., Biggs, J., Albino, F., Hill, P., & Bull, D. (2018). Application of machine learning to classification of volcanic deformation in routinely generated InSAR data. *Journal of Geophysical Research. Solid Earth*, 123(8), 6592–6606. <https://doi.org/10.1029/2018JB015911>
 - a. The study by Anantrasirichai et al. (2018) introduces a machine learning approach using convolutional neural networks (CNNs) to automate the detection of volcanic deformation in wrapped InSAR interferograms. Given the vast amount of Sentinel-1 satellite data covering over 900 volcanoes globally, the authors trained a CNN (specifically [AlexNet](#), with transfer learning from ImageNet) on historical Envisat data to classify deformation patterns. They applied this to over 30,000 Sentinel-1 interferograms, initially flagging 1,369 as potential deformation events. Expert validation confirmed 39 true positives, leading to a retrained model that reduced false positives and streamlined analysis. Despite challenges like atmospheric artifacts and subtle deformation signals, the study demonstrates the potential of CNNs to significantly reduce manual effort in global volcano monitoring while maintaining detection reliability.
2. Anantrasirichai, N., Biggs, J., Albino, F., & Bull, D. (2019). A deep learning approach to

detecting volcano deformation from satellite imagery using synthetic datasets.

Remote Sensing of Environment, 230, 111179-.

<https://doi.org/10.1016/j.rse.2019.04.032>

- a. The study by Anantrasirichai et al. (2019) presents a deep learning framework for automatically detecting volcanic deformation using satellite-based InSAR imagery. Due to the scarcity of deformation events and the dominance of atmospheric noise, training effective machine learning models on real data alone is challenging. To address this, the authors generate synthetic interferograms that combine simulated deformation patterns with modeled atmospheric noise—both stratified and turbulent. They train a convolutional neural network ([AlexNet](#)) using these synthetic datasets and then refine the model with selected real-world examples. Tested on over 30,000 Sentinel-1 interferograms, the approach significantly improves detection accuracy, reducing false positives by more than 80% compared to previous methods. The authors further enhance performance by applying atmospheric corrections to high-probability detections. Their results demonstrate that using synthetic data not only compensates for the imbalance in real training data but also enables reliable, scalable, and efficient monitoring of global volcanic activity.

Methodology

Data Collection and Preprocessing

- **Data Source**

InSAR datasets for Kīlauea and the Reykjanes Peninsula were acquired using the RADARSAT Constellation Mission (RCM), operated by the Canadian Space Agency. For each site, time series of Single Look Complex (SLC) SAR images were collected using beam modes optimized for regional coverage and coherence. The image stacks were acquired along ascending orbit tracks, with incidence angles and acquisition geometries selected to balance deformation sensitivity and coherence preservation in volcanic and coastal environments. The spatial resolution of the imagery is approximately 5 metres, with a nominal revisit time of 12 days; however, variable acquisition intervals occurred due to satellite scheduling constraints.

A fully automated InSAR processing workflow was used to generate a standard set of geospatial products for each image pair, including wrapped and unwrapped interferograms, coherence maps, and line-of-sight (LOS) displacement maps. The wrapped interferogram displays the phase difference between two radar acquisitions as cyclical color bands, where each full fringe represents a fixed interval of surface displacement. While visually informative, wrapped interferograms require further processing to quantify actual motion. To address this, the unwrapped interferogram removes the 2π phase ambiguity, converting the cyclical pattern into a continuous surface that reflects the magnitude and direction of deformation over time. Alongside these, the coherence map provides a measure of signal stability between image pairs, with values ranging from 0 (low coherence) to 1 (high coherence); high coherence indicates minimal surface change, while low coherence often reflects vegetation, snow,

or rapid surface disruption. Finally, the LOS displacement map translates the unwrapped phase data into ground displacement measurements along the satellite's line of sight, offering a more direct and physically meaningful interpretation of surface motion.

The workflow is based on [Gamma software](#) and uses [processing scripts](#) developed by Canada Centre for Mapping and Earth Observation (CCMEO). Topographic phase was removed using United States Geological Survey (USGS) [1 arc-second global Digital Elevation Model \(DEM\)](#), and definitive satellite orbits were applied to reduce orbital errors. An additional baseline refinement step was included to further improve accuracy. To reduce noise, SAR images were multi looked by a factor of two in both range and azimuth. Interferograms were unwrapped using the Minimum Cost Flow (MCF) algorithm and converted to LOS displacement maps.

- **Storage**

All data is stored in the AWS environment to enable efficient access and processing. For each Volcanic Site/Beam combination, we retrieve displacement images (`startdate_HH_enddate_HH.disp.geo.tif`) and wrapped phase images (`startdate_HH_enddate_HH.adf.wrp.geo.tif`) based on either the specified date ranges (Milestone 1) or all available images (Milestone 2). When running an AWS Batch job (Milestone 3), files are accessed directly from S3, eliminating the need for local downloads and streamlining the processing workflow.

- **Preprocessing**

To ensure compatibility between RCM InSAR data and the pre-trained convolutional neural network (CNN) architecture used in this study, spatial resolution adjustment was a critical preprocessing step. The model, based on [AlexNet](#), was originally trained on Envisat and Sentinel-1 imagery, which have specific spatial characteristics. Because CNNs are sensitive to the scale and appearance of input features, mismatches in spatial resolution can lead to poor feature recognition and degraded model performance.

High-resolution RCM data can result in deformation fringes appearing sparse or noisy, while lower resolutions risk blurring or losing essential deformation patterns. To address this, the RCM data provided by the Canadian Space Agency was resampled across a range of spatial resolutions—5 to 100 meters in Universal Transverse Mercator (UTM) coordinates (in 5 meter increments) and 0.00005 to 0.001 degrees in geographic coordinates (in 10 steps). This range allowed for the identification of an optimal resolution that balances the preservation of key deformation features with the input characteristics expected by the model.

Additionally, adjustments were made in both Universal Transverse Mercator (UTM) and geographic projections to ensure compatibility across coordinate systems. This is particularly important when working with high-resolution, region-specific RCM imagery. Overall, this preprocessing step was essential for improving the model's ability to

generalize and for minimizing false positives when applying a synthetic-data-trained CNN to real-world RCM observations.

Machine Learning Models

We selected two pre-trained CNN models based on previous studies in volcanic deformation detection using InSAR data. The models' architecture and implementation details are publicly available in the research repository: [volcano_deform_detection](#)

- #### Model 1

Model 1 is based on a convolutional neural network (CNN) architecture, specifically using the [AlexNet](#) model, to classify volcanic deformation in wrapped InSAR interferograms. [AlexNet](#) was selected for its strong performance in initial testing compared to other architectures like [ResNet50](#) and [InceptionV3](#), particularly in terms of true positive rate (TPR) and area under the curve (AUC).

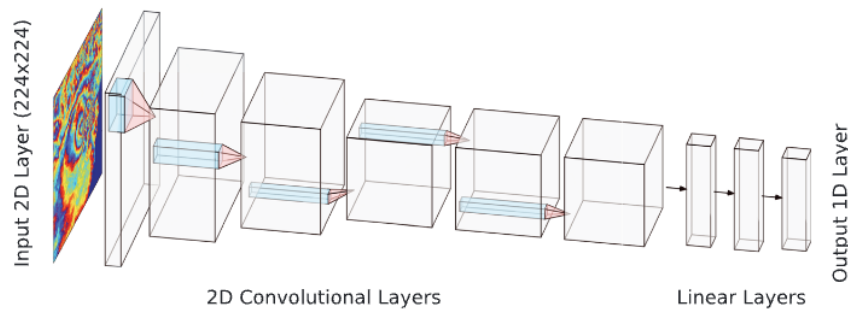


Figure 4: Structure of the AlexNet convolutional neural network adapted for InSAR input. The input image (224×224) is processed through five 2D convolutional layers with filter sizes of 11×11 , 5×5 , and three 3×3 layers, each followed by ReLU activation and max pooling (3×3). The final three layers are fully connected, each generating 4,096 features. The blue block illustrates how neighboring pixels are combined during convolution to produce a single output for the next layer.

The model takes 224×224 grayscale patches of interferograms as input, which are derived from wrapped phase values ($-\pi$ to π) scaled to 0–255. [AlexNet](#) consists of five convolutional layers followed by three fully connected layers, using ReLU activations and max pooling, with a final softmax layer to output class probabilities. To adapt the model to this geospatial task, transfer learning was applied: pretrained weights from ImageNet were used, with the final layers replaced and fine-tuned using labeled InSAR data. Only patches with strong edges (detected using the Canny edge detector) were classified, reducing noise and focusing computation on likely deformation areas.

Given the limited number of positive samples, the team used extensive data augmentation—including shifts, flips, rotations, and affine distortions—to expand the set of deformation examples to around 10,000 patches. Negative samples were randomly sampled to match this number, ensuring balanced training. Training was performed using mini-batches with a maximum of 50 epochs, and the model was trained and tested using twofold cross-validation on both Envisat and Sentinel-1 datasets.

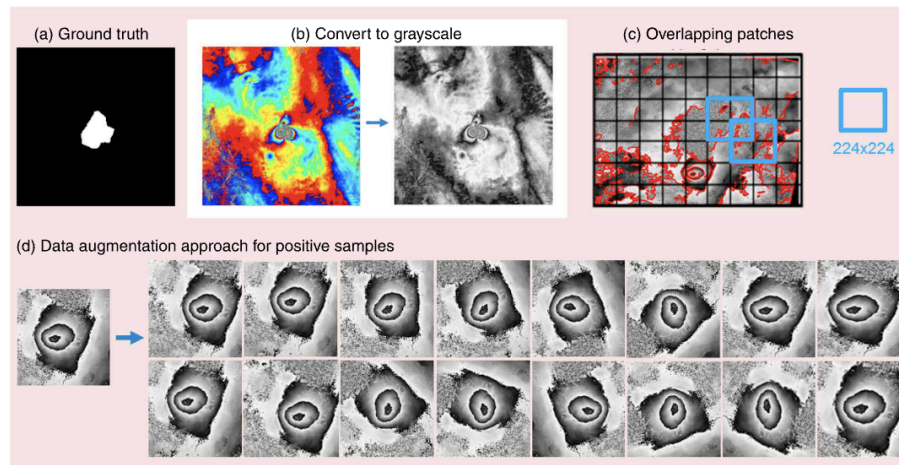


Figure 5: Data preparation workflow. (a) Expert-labeled ground truth, with white indicating positive and black indicating negative samples; (b) Phase values ranging from $-\pi$ to π are scaled to grayscale (0–255) for CNN compatibility; (c) Overlapping patches are extracted using a sliding window (e.g., 224×224 pixels) with 50% overlap—patches without edges are automatically labeled negative and excluded from training; (d) Data augmentation increases the number of positive samples through rotations, flips, distortions, and pixel shifts to balance the dataset.

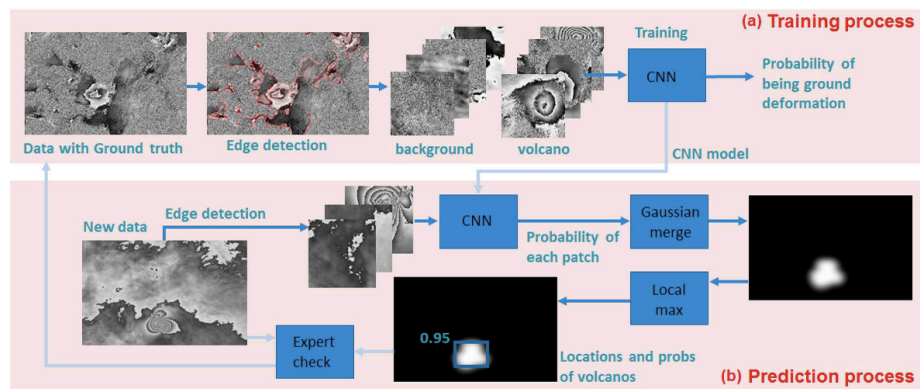


Figure 6: Overview of the proposed framework of Model 1, including (a) training and (b) prediction stages. In training, ground-truth-labeled data (1 = deformation, 0 = no deformation) undergo edge detection to focus on regions with significant phase changes. These regions are divided into positive and negative patches and used to train the CNN. During prediction, a new interferogram is split into overlapping patches; those with strong edges are evaluated by the trained CNN to estimate the probability P of deformation. Probabilities are merged using Gaussian weighting, and the highest probability ($P_{\max}$) and its location are identified to support alert generation. An expert then reviews the results, and confirmed positives are used to further retrain the CNN.

[AlexNet](#) consistently outperformed support vector machines (SVMs) and other CNN architectures across accuracy, True Positive Rate (TPR), and True Negative Rate (TNR) metrics. Once trained, the model was applied to a global dataset of over 30,000 interferograms, where it classified overlapping patches, merged probabilities using Gaussian weighting, and flagged high-probability detections for expert review. This expert feedback loop was then used to retrain and refine the model further. All training and inference tasks were performed on the University of Bristol's high-performance computing cluster, with training taking approximately 38 hours for the initial model and 26 hours for retraining, while inference on a single interferogram took about 1.5 seconds.

This pipeline demonstrated that CNNs, when paired with edge-based filtering and transfer learning, can significantly reduce manual effort in detecting volcanic deformation from large-scale satellite data.

- **Model 2**

The machine learning framework in this study is based on a deep convolutional neural network (CNN), specifically [AlexNet](#), which was selected due to its superior performance in earlier volcanic deformation classification tasks compared to other pre-trained architectures. [AlexNet](#) consists of five convolutional layers, three of which are followed by max pooling layers, and ends with three fully connected layers, including a final softmax layer for classification (Figure 4). Non-linear activation is introduced using ReLU after every convolutional and fully connected layer.

The CNN was implemented using a transfer learning approach. Rather than training the entire model from scratch, the researchers fine-tuned a pre-trained [AlexNet](#) originally trained on ImageNet (which contains millions of natural images). For this application, the final fully connected and softmax layers were replaced to suit a binary classification task (deformation vs. non-deformation) or multi-class models (including combinations of deformation, stratified atmosphere, and turbulent atmosphere). Only the final layers were trained with a significantly higher learning rate, while the earlier layers retained the features learned from general image classification.

To overcome the severe class imbalance in real InSAR datasets—where <0.2% of images show actual deformation—synthetic interferograms that mimicked the physical characteristics of InSAR signals were generated. These were created by combining three components:

- [Deformation](#) (D) modeled using analytical sources like Mogi (point pressure), penny-shaped cracks (sills), and Okada dislocations (earthquakes, dykes).
- **Stratified Atmospheric Delays** (S), simulated using Generic Atmospheric Correction Online Service (GACOS) weather data.
- **Turbulent atmospheric noise** (T), statistically generated from empirical variogram models using Monte Carlo sampling.

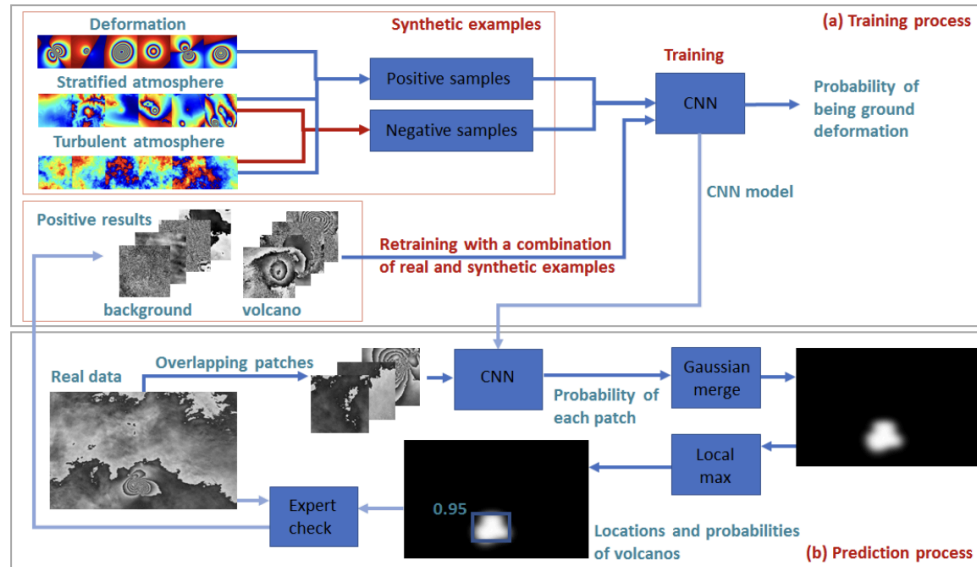


Figure 7: The proposed framework of Model 2 includes (a) the training phase and (b) the prediction phase. Initially, synthetic data are used to train the CNN and create a baseline model. In the prediction phase, patches from new interferograms are evaluated by the CNN, producing deformation probabilities (P) that are merged using Gaussian weighting. An expert reviews the results, and both true and false positives are incorporated into retraining the CNN using a mix of real and synthetic data to improve performance.

Each component was used individually and in various combinations (e.g., D+S+T, S+T, etc.) to form large, balanced training datasets. They trained multiple models with:

- 2-class (deformation vs. non-deformation),
- 3-class (D vs. S vs. T and variants), and
- 91-class models, where different weights of deformation, stratified, and turbulent components were combined and classified as a regression problem (estimating $\alpha D + \beta S + \gamma T$ weightings).

Input images were converted to grayscale and normalized, then split into 224x224 pixel patches—the input size required by [AlexNet](#). During testing, each InSAR interferogram was subdivided into overlapping patches. Canny edge detection was used to identify phase jumps and exclude low-information regions, which sped up inference.

Training was conducted on a high-performance GPU cluster, with models trained for up to 50 epochs using mini-batch gradient descent (batch size = 100). Early stopping was used to prevent overfitting, and performance was evaluated on both synthetic and real datasets. The models were retrained using a combination of synthetic and selected real examples, with significant gains in positive predictive value (up to 82%) and a large reduction in false positives.

Finally, atmospheric correction was applied to the top-ranked positive detections using GACOS products to further improve classification and eliminate false positives caused by stratified atmospheric signals.

- **Comparison Between Model 1 and Model 2**

Although both models are based on the AlexNet CNN architecture, utilize transfer learning for classifying deformation in InSAR interferograms, and utilize Canny edge detection to filter out low-information patches, they differ significantly in their data sources, training strategies, and classification objectives.

	Model 1	Model 2
Training Data	Model 1 relies solely on real InSAR interferograms . Positive samples are manually labeled based on expert interpretation, while negative samples are extracted from non-deforming regions. Data augmentation (rotations, flips, distortions) is heavily used to expand the positive sample pool.	Model 2 uses synthetic interferograms to address the class imbalance problem. These are generated by combining simulated deformation sources (e.g., Mogi, Okada), stratified atmospheric delays (from GACOS), and turbulent noise (Monte Carlo simulations). Real interferograms are later incorporated for retraining and validation.
Classification Objectives	Model 1 performs binary classification (deformation vs. non-deformation), focused on detecting deformation signals in real interferograms.	Model 2 supports multiple classification schemes: • 2-class: deformation vs. non-deformation, • 3-class: deformation, stratified atmosphere, turbulent atmosphere, • 91-class: regression-based classification predicting weighted mixtures of D, S, and T components.
Edge Detection and Patch Selection	Model 1 applies Canny edge detection primarily for reducing computation on real data.	Model 2 integrates Canny edge detection into both synthetic and real data workflows, using it to enhance training efficiency and focus on informative regions.
Retraining Strategy	Model 1 follows an expert-in-the-loop approach where confirmed detections are used to continually improve the model.	Model 2 also uses expert-reviewed detections but uniquely combines them with synthetic data to retrain models, improving generalization and reducing false positives.

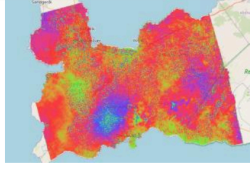
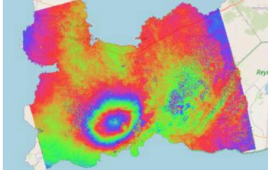
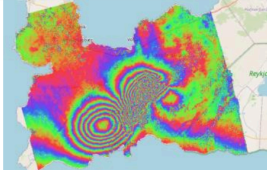
Atmospheric Correction	Model 1 does not apply atmospheric correction in its inference pipeline.	Model 2 incorporates GACOS-based correction during post-processing to eliminate false positives from stratified atmospheric effects.
Objective and Scope	Model 1 is optimized for high true positive detection in operational monitoring of real-world deformation signals.	Model 2 aims for broader generalization and interpretability , allowing it to distinguish between geophysical and atmospheric components and simulate a wide range of scenarios.

Implementation

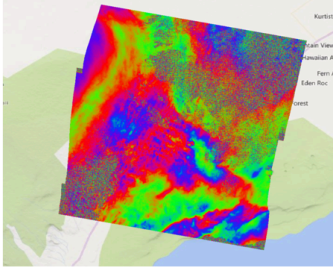
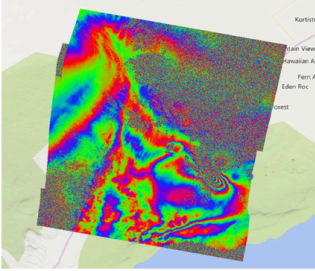
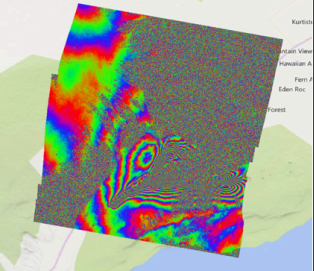
To explore the practical applications of InSAR and machine learning in volcanic monitoring, this project focused on two case study sites: Kīlauea in Hawai'i and the Reykjanes Peninsula in Iceland. These sites were selected due to their recent and ongoing volcanic activity, as well as the availability of high-quality satellite data.

Kīlauea is one of the most active volcanoes in the world. Its most recent eruption began in December 2024 and is still ongoing, with multiple episodes of lava fountaining. All activity remains within Hawai'i Volcanoes National Park. Meanwhile, the Reykjanes Peninsula has seen a series of eruptions since 2021, marking the start of a new active period after centuries of dormancy. The most recent eruption began in late 2024 near Grindavík and continues to show signs of unrest.

We focus on different levels of deformation (Low, Medium, High) and pick out an interferogram for each level for each site. Visually, this means that the closer the circle fringes are together, the higher the level of deformation.

Image			
Start Date	2024/06/29	2024/04/22	2024/07/15
End Date	2024/07/03	2024/05/04	2024/09/01
Deformation	Low	Medium	High

Reykjanes/5M4

Image			
Start Date	2024/08/02	2024/08/02	2024/05/18
End Date	2024/08/10	2024/08/22	2024/08/10
Deformation	Low	Medium	High

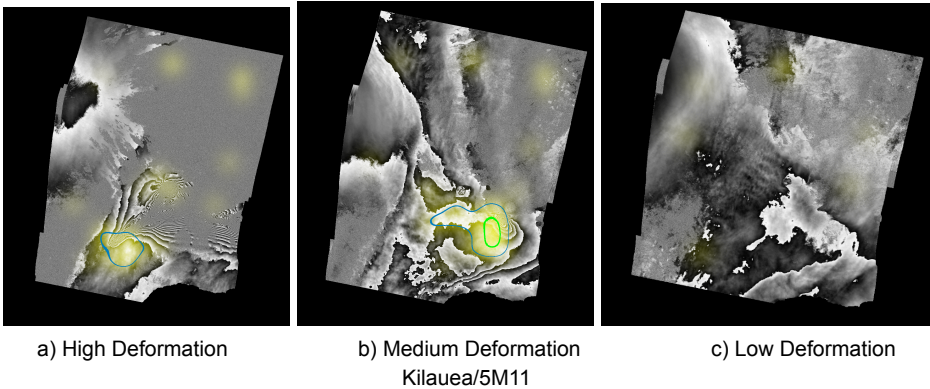
Kilauea/5M11

Figure 8: Selected interferogram examples for Reykjanes (5M4 beam) and Kilauea (5M11 beam), illustrating varying levels of deformation: low, medium, and high.

To determine the optimal image resolution for our volcanic deformation detection models, we ran the `run_volcano_deformation_detection` function across a variety of configurations. This involved testing combinations of Model 1 and Model 2, both Universal Transverse Mercator (UTM) and geographic coordinate systems, and multiple image resolutions. The objective was to assess how each of these settings influenced the AI model's outputs.

For each configuration, we recorded several evaluation metrics, including the maximum pixel probability (the model's highest confidence in detecting deformation at any point in the image), and the percentage of pixels exceeding 50% and 80% probability thresholds. Additionally, we conducted qualitative assessments of the AI-generated outputs. To keep the analysis focused and manageable, we selected one representative date per deformation level (high, medium, low) for each site.

Each run produced an RGB probability map alongside the numerical metrics. In these maps, green circles indicate regions where pixel probabilities exceeded 80%, blue circles highlight areas exceeding the 50% threshold, and yellowish regions represent areas with moderate model confidence (below 50%).



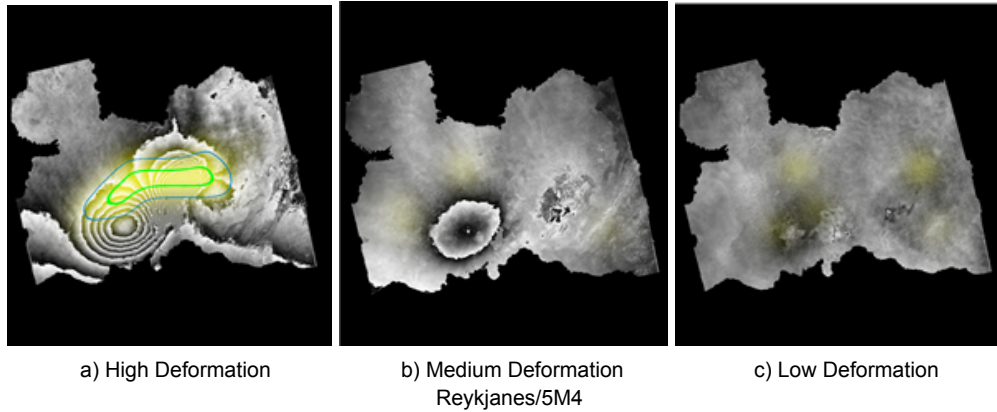


Figure 9: AI-generated RGB probability maps using the optimal configuration (Model 2, 40 m resolution, and UTM coordinates), shown for each deformation level: high, medium, and low.

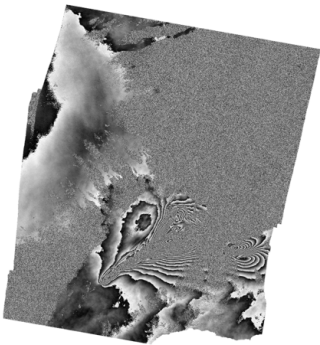
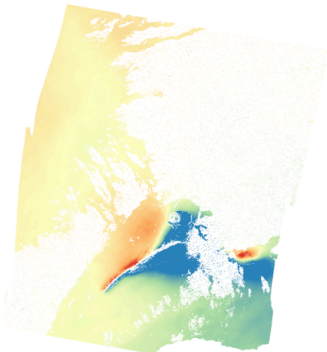
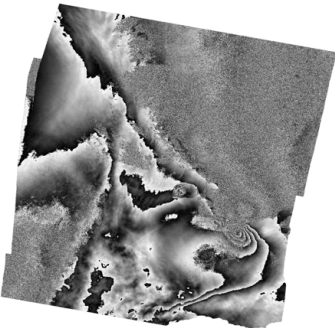
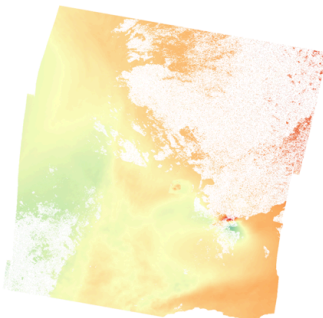
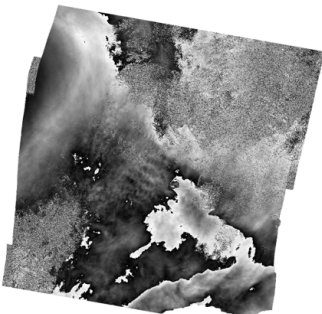
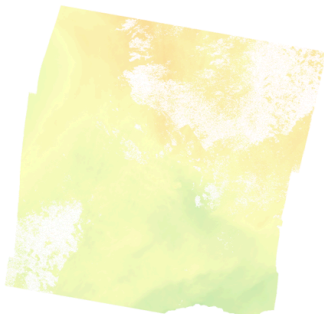
After identifying the optimal setup—Model 2, 40 m resolution, and Universal Transverse Mercator (UTM) coordinate system—we process all available images for each site using the `run_volcano_deformation_detection_single` function.

We compute values like Minimum Value, Maximum Value, Absolute Maximum Value, Range, 25th Percentile, 50th Percentile (Median), 75th Percentile, Normalized Displacement (50% Threshold), Normalized Displacement (80% Threshold), [AI] Max Probability, [AI] Percent Above 50%, and [AI] Percent Above 80% for each Start Date–End Date pair that has both displacement and wrapped phase images.

The [AI] outputs refer to the values produced by running the AI model on the wrapped phase image, as done in the previous step. The normalized displacement refers to calculating the average displacement within the regions where the AI model is at least 50% or 80% confident, divided by the maximum absolute displacement value in the full scene. This provides a confidence-weighted measure of how strong the ground motion is in the areas flagged by the model, relative to the strongest displacement observed.

Deformation	Date Range	wrp	disp
High	2024/07/15- 2024/09/01		
Medium	2024/04/22- 2024/05/04		
Low	2024/06/29- 2024/07/03		

Reykjanes/5M4

Deformation	Date Range	<u>wrp</u>	<u>disp</u>
High	2024/05/18-2024/08/10		
Medium	2024/08/02-2024/08/22		
Low	2024/08/02-2024/08/10		

Kilauea/5M11

Value	Color	Label
-0.15		-0.1500
-0.075		-0.0750
0		0.0000
0.075		0.0750
0.15		0.1500

Displacement Map Legend

Figure 10: Wrapped phase (grayscale) and displacement images for Reykjanes (5M4 beam) and Kīlauea (5M11 beam), along with a legend quantifying displacement levels for each interferogram. The wrapped phase fringes can be visually compared to the displacement maps to assess how fringe density relates to the magnitude of ground deformation.

The output from each site was saved as a CSV file, resulting in a substantial dataset that was imported into Google Sheets for visualization and trend analysis.

For visualization, we plotted the dependent variables—[AI] Max Probability, Normalized Displacement (50% Threshold), and Normalized Displacement (80% Threshold)—against a range of statistical descriptors used as independent variables: Minimum Value, Maximum Value, Absolute Maximum Value, Range, 25th Percentile, 50th Percentile (Median), and 75th Percentile. The [raw data](#) and [resulting plots](#) are available.

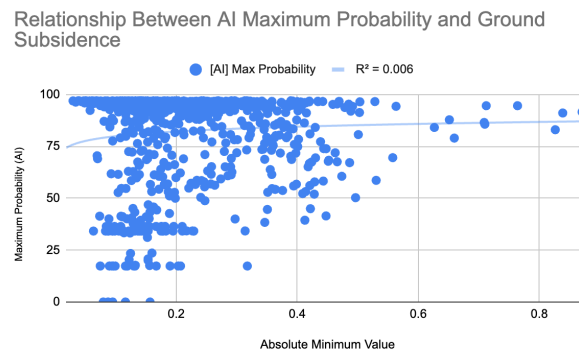


Figure 11: Example visualization from Milestone 2, showing a plot of maximum AI-predicted probability (dependent variable) versus the absolute minimum displacement value (independent variable) for Reykjanes/5M4. A logarithmic trend line is fitted to illustrate the relationship.

Milestone 3 focused on automation and scalability—key priorities for transitioning the model from a research prototype into a deployable, production-ready system. One of the main goals was to eliminate dependencies on local storage and manual intervention. To achieve this, we modified the project’s Dockerfile and restructured the codebase so that the entire pipeline could directly access and process images stored in Amazon S3. This shift from local file access to cloud-based storage not only enhances scalability but also enables consistent, centralized data handling across different computing environments.

By configuring the system to reference S3, the pipeline can now be executed in a single streamlined run, automatically reproducing the full data generation process from Milestone 2—from image ingestion to model inference and output generation. This reproducibility is critical for ensuring consistency in model evaluations and for supporting large-scale batch processing.

To further support automation, the setup is currently being integrated with AWS Batch, which will allow for distributed, on-demand processing of large volumes of SAR data. This integration is a key step toward operationalizing the system, enabling it to run continuously or on a schedule without manual triggering. While integration is nearly complete, we are still in the

process of resolving access and permission issues related to S3, which are currently the main barrier to full automation.

Evaluation Metrics

In Milestone 1, the evaluation of model performance for detecting volcanic deformation via RGB probability maps (see Figure 9 for an example) combined both qualitative and quantitative approaches to capture statistical accuracy, visual interpretability, and operational viability.

Each model output was assessed by measuring the maximum pixel probability, representing the model's highest confidence in detecting deformation at any point in the image. In addition, we computed the percentage of pixels exceeding 50% and 80% probability thresholds, which provided insight into the spatial extent and certainty of detections. High percentages above these thresholds typically suggest strong and spatially coherent signals across the deformation fringe. The 80% threshold was particularly important for identifying high-confidence detections (represented green circles in RGB maps), while the 50% threshold highlighted moderate confidence areas (represented blue circles in RGB maps). The yellowing represented areas with a little bit of confidence but less than 50%.

Additionally, RGB probability maps were visually inspected to determine how well the model predictions aligned with the known deformation patterns. Key qualitative indicators included:

- The presence of concentrated color features (e.g., yellowing in the deformation fringe), indicating accurate gradient capture.
- The occurrence of randomly scattered artifacts, especially along image edges, often signifying false positives and reduced prediction quality.
- The overall clarity and spatial precision of the detected deformation zone.

To ensure fair comparison across configurations, all models were tested using a consistent range of spatial resolutions: Universal Transverse Mercator (UTM) resolutions from 5m to 100m, and geographic resolutions from 0.00005° to 0.001° . RGB outputs were only generated when the maximum probability exceeded 10%, ensuring the visualizations represented meaningful detection attempts.

From this analysis, we found that Model 2, when combined with Universal Transverse Mercator (UTM) coordinates, consistently produced outputs that were more interpretable and spatially coherent. Although both configurations achieved similar peak probability values, visual inspection highlighted a key distinction: the output from Model 1 contained noticeably more random or misplaced blue circles, along with a greater prevalence of yellow regions—indicators of false positives—not seen in the Model 2 results (Figure 9).

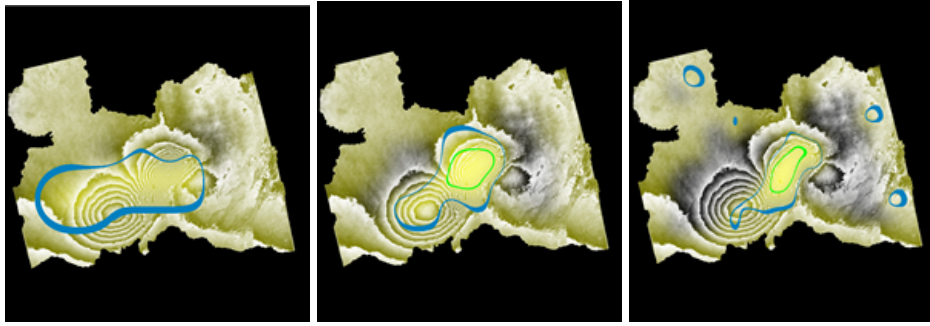


Figure 12: AI output for Reykjanes high deformation using Model 1 with UTM coordinates, shown at resolutions of 45, 35, and 25 meters (left to right). At 45 m, the model fails to highlight the correct fringe areas. The 35 m output shows slight improvement but still displays edge-related yellowing, while the 25 m resolution introduces random, inaccurate blue circles unrelated to actual deformation.

This underscores the importance of incorporating both numerical metrics and visual analysis to understand not just prediction strength, but also spatial reliability. Together, these methods formed a robust framework for evaluating model outputs across different configurations and helped identify combinations that balance sensitivity with precision in detecting volcanic deformation signals.

Building on those insights, Milestone 2 involved a more quantitative approach. We analyzed the CSV outputs generated for each site, comparing detection results across time and deformation levels. This allowed us to identify patterns in model behavior, including site-specific differences in detection sensitivity and prediction stability.

To deepen our analysis, we plotted a [series of graphs](#) using key statistical descriptors as independent variables—including minimum, maximum, absolute maximum, range, and the 25th, 50th (median), and 75th percentiles—against various dependent variables: maximum predicted probability, normalized displacement at the 50% threshold, and normalized displacement at the 80% threshold. This enabled us to explore how changes in deformation patterns influenced model outputs across different volcanic contexts.

We also implemented a binned evaluation of predicted probabilities as a function of InSAR displacement magnitude. Model outputs were grouped into fixed displacement intervals, and the distribution of predicted probabilities within each bin was analyzed. This allowed us to assess how model confidence varied across the deformation spectrum. To summarize the overall detection trend, a sigmoid function was fitted to the median probability values across bins, offering an interpretable estimate of the displacement threshold at which detections become more consistent. The accompanying box plots provided a clear visual representation of variability, interquartile range, and outliers within each bin, giving insight into the stability and spread of model predictions.

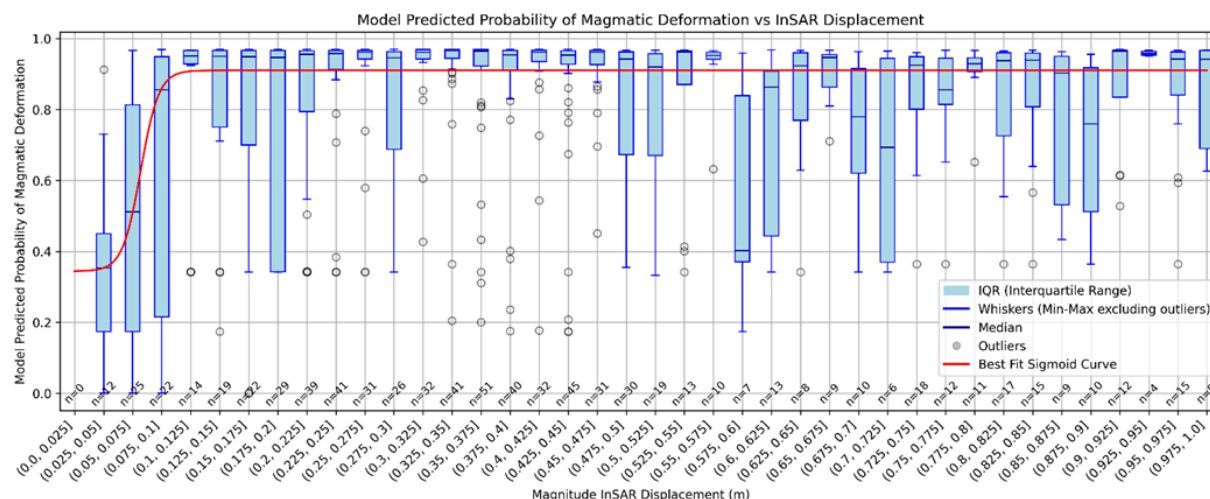


Figure 13: Distribution of model-predicted probabilities of magmatic deformation across binned InSAR displacement magnitudes. Each box represents the interquartile range of predicted probabilities for a given displacement interval, with the median shown as a horizontal line. Outliers are indicated as individual points. A sigmoid curve is overlaid to illustrate the overall detection trend, highlighting the displacement threshold at which the model begins to consistently assign higher probabilities. This visualization provides a quantitative view of model sensitivity to deformation magnitude and supports threshold estimation for reliable detection.

This combined evaluation approach provided a robust framework to characterize model sensitivity and interpret its behavior across varying levels of deformation, setting the stage for more detailed comparisons between volcanic sites and event types.

For Milestone 3, the primary evaluation focus shifted from model accuracy to operational efficiency, with an emphasis on assessing the performance and reliability of the automated pipeline. Key metrics include the automation success rate, which tracks the percentage of batch jobs that complete without manual intervention, and average runtime per job, which reflects the system's processing efficiency. Pipeline reproducibility is also critical, ensuring that identical inputs yield consistent outputs across runs. As the system relies heavily on cloud storage, S3 access latency and throughput are being monitored to identify any bottlenecks in data retrieval or output writing. Additionally, resource utilization metrics, such as CPU and memory usage, are evaluated to guide optimization of Docker container settings and AWS Batch configurations. Error rates and failure logs provide insight into the system's robustness and help identify recurring issues—such as the current S3 access challenges—that require resolution. Finally, scalability testing is ongoing to determine how well the pipeline performs when processing multiple sites or time steps in parallel, which is essential for enabling large-scale deployment.

Results

Findings and Analysis

- **Model Testing on Known Deformation Events**

The two pre-trained convolutional neural network (CNN) models were evaluated using RCM InSAR data from two volcanic sites with well-documented deformation events: Kīlauea (Hawai'i) and Reykjanes (Iceland). These sites were chosen to represent a

range of deformation intensities and styles, enabling a more comprehensive evaluation of model behavior under different conditions.

Model 1 consistently outperformed Model 2 in terms of maximum pixel probability, especially in high and medium deformation scenarios. This suggests that Model 1 is more sensitive to strong deformation signals, making it suitable for flagging high-confidence events. However, this came at a cost: Model 1 tended to produce less spatially coherent detections, occasionally highlighting false positives or fragmented regions.

In contrast, Model 2 delivered more spatially accurate predictions, with detected regions better aligning with ground-truth deformation zones. While its maximum probability scores were generally lower than those of Model 1, the predictions were more consistent and interpretable, particularly in low-deformation cases. This trade-off between sensitivity and spatial precision may inform model selection depending on operational goals—for example, whether high recall or spatial localization is prioritized.

Another key finding was the effect of the coordinate reference system on model output. Across all configurations and deformation levels, using the Universal Transverse Mercator (UTM) system resulted in cleaner outputs with fewer artifacts and higher average confidence scores, compared to the traditional geographic grid. The Universal Transverse Mercator (UTM) system preserved spatial structure more effectively, likely due to its consistent scale and reduced distortion, particularly important for convolutional operations in the CNN.

Together, these results underscore the importance of both model selection and coordinate system configuration in optimizing deformation detection performance. While Model 1 may be better for initial anomaly detection, Model 2 provides more reliable spatial context, and Universal Transverse Mercator (UTM) coordinates enhance both models' output quality.

- **Sensitivity Analysis**

Kīlauea required lower levels of ground deformation—approximately 0.03 to 0.06 normalized displacement—for the model to register positive detections, whereas the Reykjanes Peninsula typically needed higher deformation levels, between 0.08 and 0.22. Despite requiring stronger signals, the model performed better at Reykjanes, achieving higher prediction accuracy (82–89%) and exhibiting more consistent patterns. This improved performance is likely due to the larger dataset available for Reykjanes, which provided better statistical grounding for the model.

In contrast, predictions at Kīlauea were more variable, with a wider range of deformation values and greater uncertainty, reflected in lower accuracy (57–73%). This inconsistency may be partly attributed to atmospheric effects: Kīlauea, located in a tropical climate, is more prone to atmospheric instability, leading to stronger phase change signals from moisture and turbulence. These atmospheric signals can mimic or

obscure true deformation, increasing the model's difficulty in making consistent predictions. Reykjanes, with its more stable northern climate, suffers less from such phase interference, enabling clearer interpretation of InSAR data.

Overall model recall at the 80% detection threshold remained low (2–5%), indicating that the detection strategy is conservative—prioritizing precision over recall to reduce false positives.

To assess detection sensitivity, plots of normalized displacement percentages above probability thresholds were analyzed with caution, as they can sometimes obscure actual deformation measurements. While some displacement bins display clear trends or threshold-like behavior that may be informative, others resemble random noise and may require further refinement.

To quantify sensitivity more robustly, a sigmoid curve was fitted to the median probability values binned by displacement. A probability exceeding 50% was interpreted as a positive identification of unrest. The first bin to exceed this threshold falls between 5 cm and 7.5 cm of displacement (Figure 13), indicating that the model's detection sensitivity lies within this range. For a conservative estimate, the detection threshold can be reported as approximately 7.5 cm.

- **Workflow Implementation**

To support near real-time processing and large-scale deployment of volcanic deformation detection, a scalable inference workflow was developed using Docker containers and AWS Batch. This architecture was designed to decouple the model inference process from local computing environments, enabling flexible, cloud-based execution across a distributed infrastructure. At its core, the pipeline automates the full data flow—from pulling interferograms (IFGs) from cloud storage, to preprocessing, running AI inference, and exporting results for analysis.

Using AWS Batch, the system demonstrated strong scalability, successfully processing multiple IFGs concurrently across different volcanic sites and time periods. The ability to parallelize jobs allows for significant reductions in total processing time, opening the door to continuous monitoring of deformation signals as new satellite data becomes available.

The workflow is containerized using Docker, ensuring consistent execution environments and easier reproducibility across development, testing, and production phases. This containerization also makes it easier to deploy updates to the model or preprocessing logic without modifying the underlying infrastructure.

Despite these successes, a key challenge remains in fully replicating the local file access logic used in earlier stages of development within the AWS S3-based cloud environment. The transition from local disk I/O to cloud-native data handling introduces complexities, such as managing S3 permissions, optimizing file access patterns, and ensuring seamless integration with tools that expect file-based input. These challenges

are currently the focus of optimization efforts, with particular attention to ensuring that S3 latency and file availability do not introduce bottlenecks or reduce overall reliability of the workflow.

Once these issues are resolved, the system will be capable of fully automated, on-demand deformation detection at scale, forming the backbone for operational deployment of AI-assisted volcanic monitoring.

Discussion

Implications, Limitations, and Future Work

The application of pre-trained CNNs to RCM InSAR data demonstrated promising detection performance, especially in high-deformation scenarios. However, sensitivity analysis revealed a consistent trade-off between recall and spatial coherence. While predictions at the Reykjanes site were more accurate and stable due to a larger, more uniform dataset, predictions at Kilauea were less consistent, likely due to atmospheric interference and a smaller sample size. These results suggest that model effectiveness is influenced by both deformation magnitude and local environmental conditions.

Although detection thresholds were established through sigmoid fitting—approximately 7.5 cm of displacement for confident identification—the recall at 80% probability remained low (2–5%), indicating the current models are conservative and optimized for precision. Improving sensitivity without increasing false positives remains an open challenge.

The AWS Batch-based pipeline successfully automated model inference and supported batch processing across multiple interferograms. However, full automation is limited by ongoing S3 access and permission issues. Addressing these barriers will be essential for enabling continuous, large-scale deployment. Future work should include expanding the training dataset to include additional volcanic sites, incorporating atmospheric corrections during inference, and improving model generalization across diverse volcanic settings.

Conclusion

This project evaluated the use of pre-trained convolutional neural networks (CNNs) for detecting volcanic unrest in RCM InSAR data. Through testing and sensitivity analysis, several key findings emerged:

- Model 2 produced more spatially coherent predictions than Model 1, especially when paired with Universal Transverse Mercator (UTM) coordinates.
- Detection sensitivity varied by site, with Reykjanes requiring greater deformation but yielding higher prediction accuracy.
- Model sensitivity was limited in low-deformation scenarios, as indicated by low recall at high thresholds.
- A scalable, cloud-based workflow was developed using AWS Batch and Docker, allowing for automated inference across multiple datasets, although full S3 integration

remains pending. This system lays the groundwork for operational volcanic monitoring pipelines.

These findings support the feasibility of using machine learning for large-scale volcanic unrest detection and highlight areas for further improvement.

Final Thoughts

This work demonstrates the potential of combining pre-trained CNNs with high-resolution RCM InSAR data to automate volcanic deformation detection. The sensitivity analysis, case study evaluations, and deployment of a cloud-based pipeline provide a strong foundation for operational use. While current limitations in sensitivity and cloud integration exist, the system offers a scalable approach for real-time volcanic monitoring, especially in regions with limited ground instrumentation. Continued development will further enhance its reliability and expand its utility for global geohazard applications.

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Appendices

- [Github Repository](#)
- [Milestone 1 Demo: Literature Review, Data Acquisition & Baseline Model Implementation](#)
- [Milestone 2 Demo: Sensitivity Analysis & Model Optimization](#)
- [Milestone 3 Demo: Workflow Development](#)
-  Initial Proposal & Milestones:
-  Milestone 1: AI Output Results
-  Milestone 2: Statistical Charts
-  milestone 2: csv output data
-  Final Presentation